

A Report on

Design and Analysis of Trapezoid Ring Microstrip Patch Antenna For Lower Band 5G Application

For

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In

Electronics &a Telecommunication Engineering

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CERTIFICATE

This is to certify that the project entitled **Trapezoid Ring Microstrip Patch Antenna**
is a bonafide work of

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Submitted to the University of Mumbai in partial fulfillment of the requirement
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ABSTRACT

This report presents the design and analysis of a trapezoid ring microstrip patch antenna for lower band 5G applications centered at 3.5 GHz. The antenna is designed on an FR4 substrate with a dielectric constant of 4.3 and thickness of 1.6 mm, using a $50\ \Omega$ microstrip line feed for proper impedance matching. Simulation is carried out using CST Studio Suite 2017 over a frequency range of 2.4–4.6 GHz.

The proposed antenna achieves a resonant frequency of 3.492 GHz with a return loss of $-12.664\ \text{dB}$ and a VSWR of 1.607, indicating acceptable impedance matching. The radiation characteristics show a broadside pattern with a directivity of 6.77 dBi and a beamwidth of 96.6° . These results confirm that the antenna operates efficiently within the targeted 5G frequency band (3.4–3.6 GHz).

Due to its compact size, low cost, and ease of fabrication, the designed antenna is suitable for applications such as 5G communication systems, IoT devices, and wireless networks. The study demonstrates that trapezoidal modifications in patch geometry can effectively enhance antenna performance for modern wireless applications.

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Design and Analysis of Trapezoid Ring Microstrip Patch Antenna

For Lower Band 5G Application

Simulation performed using CST Studio Suite 2017 | Frequency Range: 2.4 – 4.6 GHz
Substrate: FR4 ($\epsilon_r = 4.3$, $\delta = 0.025$, $h = 1.6$ mm) | Feed: 50 Ω Microstrip Line

I. Introduction

The rapid advancement of fifth-generation (5G) wireless communication technology necessitates the development of compact, efficient, and cost-effective antenna designs. The lower 5G frequency band, spanning 3.4 GHz to 3.6 GHz, is a primary band allocated for 5G deployment in India and several other countries. Microstrip patch antennas are widely preferred for such applications due to their low profile, light weight, ease of fabrication, and compatibility with printed circuit board (PCB) technology.

In this report, a Trapezoid Ring Microstrip Patch Antenna is designed, simulated, and analyzed for lower band 5G operation centered at 3.5 GHz. The antenna is built on an FR4 dielectric substrate with a relative permittivity (ϵ_r) of 4.3 and loss tangent (δ) of 0.025. A microstrip line feed with a patch insertion technique is employed to achieve proper impedance matching of 50 Ω . The simulation and analysis were carried out using CST Studio Suite 2017.

II. Antenna Design

2.1 Design Methodology

The antenna dimensions are derived using the transmission line model for rectangular microstrip patch antennas. The width W of the radiating element is computed using the formulation by Bahl and Bhartia:

$$W = c / [2f_0 \times \sqrt{((\epsilon_r + 1)/2)}]$$

where c is the speed of light, f_0 is the resonant frequency, and ϵ_r is the dielectric constant of the substrate. The effective dielectric constant ϵ_{reff} is given by Balanis as:

$$\epsilon_{\text{reff}} = (\epsilon_r + 1)/2 + (\epsilon_r - 1)/2 \times [1 + 12h/W]^{(-1/2)}$$

The effective electrical length L_{eff} is then obtained as: $L_{\text{eff}} = c / (2f_0\sqrt{\epsilon_{\text{reff}}})$. The trapezoid geometry is derived by modifying the rectangular patch to redirect current paths and improve matching at 3.5 GHz.

2.2 Design Parameters

The complete set of design parameters used for the trapezoid ring microstrip patch antenna is listed in Table 1 below.

SR. NO.	Parameter Name	Designed Value
1	Resonant Frequency, F_0	3.5 GHz
2	Dielectric Constant, E_r	4.3
3	Trapezoid Patch Length, L	16 mm
4	Trapezoid Patch Lower Width, W	30 mm
5	Trapezoid Patch Upper Width, W_a	20 mm
6	Substrate & Ground Length, L_g	32 mm
7	Substrate & Ground Width, W_g	50 mm
8	Length of Inner Trapezoid, L_b	2 mm
9	Upper Width of Inner Trapezoid, W_b	2 mm
10	Lower Width of Inner Trapezoid, W_c	4 mm
11	Length of Feed Line, F_h	8 mm
12	Length of Insert Feed, F_i	4.8 mm
13	Width of Feed Line, F_w	2.748 mm

Table 1. Design parameters of the Trapezoid Ring Microstrip Patch Antenna

2.3 Antenna Structure (3D Model)

The 3D model of the antenna as rendered in CST Studio Suite 2017 is shown in Figure 1. The brown layer represents the FR4 substrate with the copper ground plane, while the yellow trapezoidal structure represents the radiating patch with its inner slot and microstrip feed line.

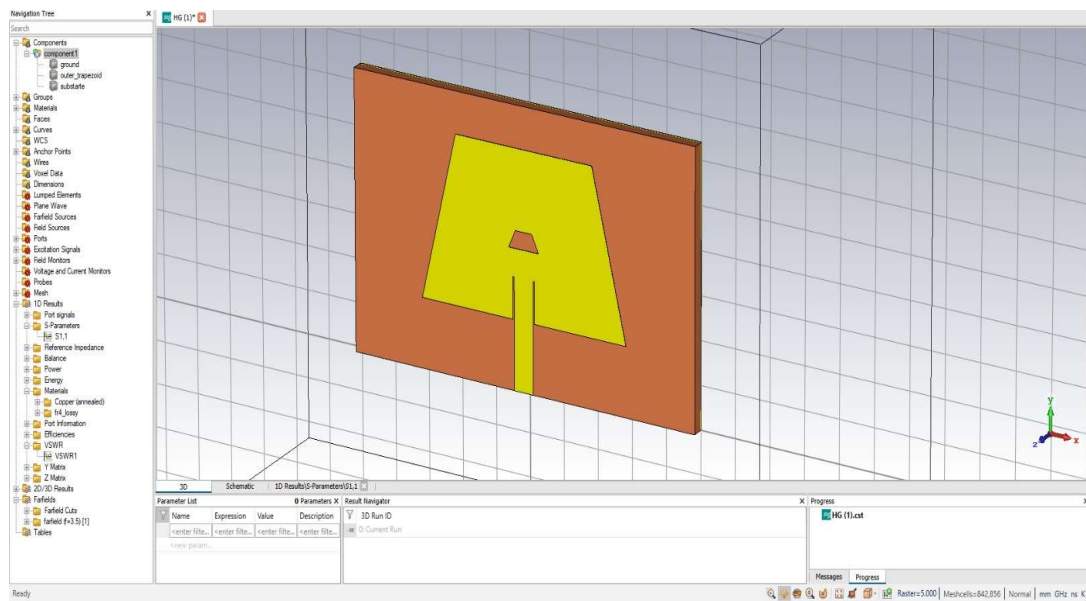


Figure 1. 3D Model of the Trapezoid Ring Microstrip Patch Antenna (CST Studio Suite 2017)

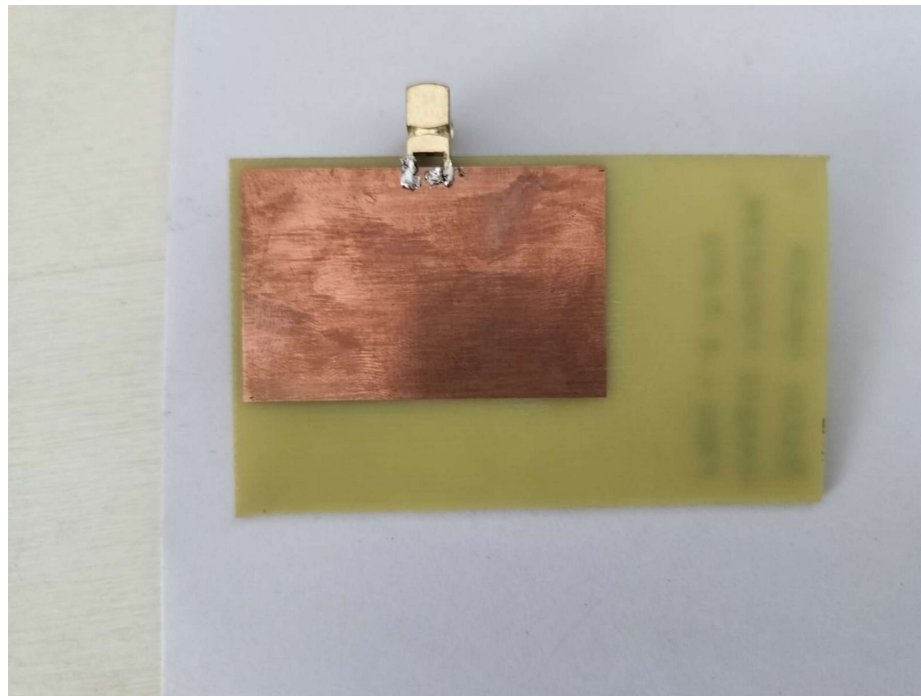


Figure A. Measurement setup – VNA S-parameter measurement of the fabricated antenna

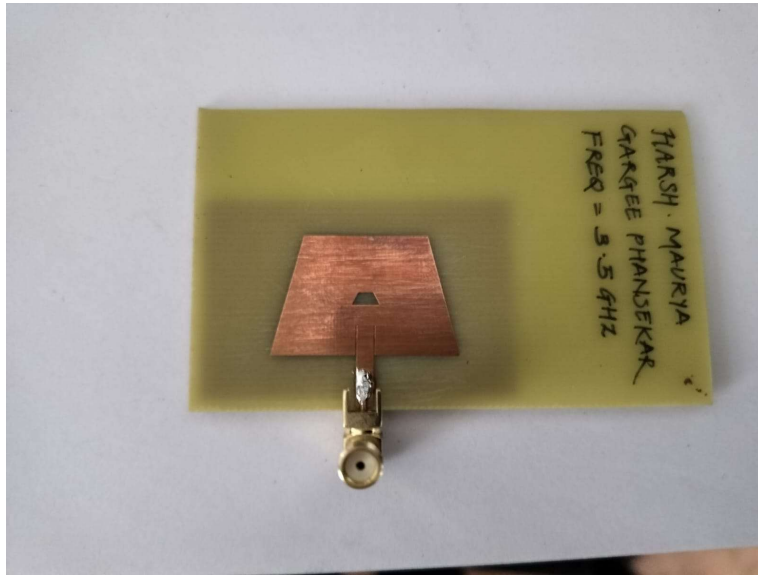


Figure B. Front view – Trapezoid copper patch with inner slot and SMA connector (Freq = 3.5 GHz)

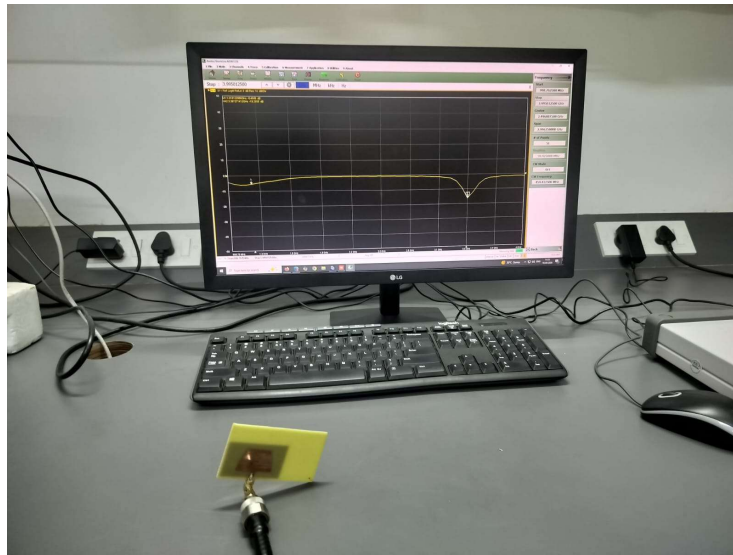


Figure C. Rear view – Ground plane of the FR4 substrate

III. Simulation Results

3.1 S-Parameters (Return Loss)

The S-parameter analysis provides the reflection coefficient ($S_{1,1}$) of the antenna as a function of frequency. A value of $S_{1,1}$ below -10 dB is the standard criterion for acceptable impedance matching. As shown in Figure 2, the antenna exhibits a minimum return loss of -12.664 dB at a resonant frequency of 3.492 GHz, confirming operation within the lower 5G frequency band (3.4 – 3.6 GHz).

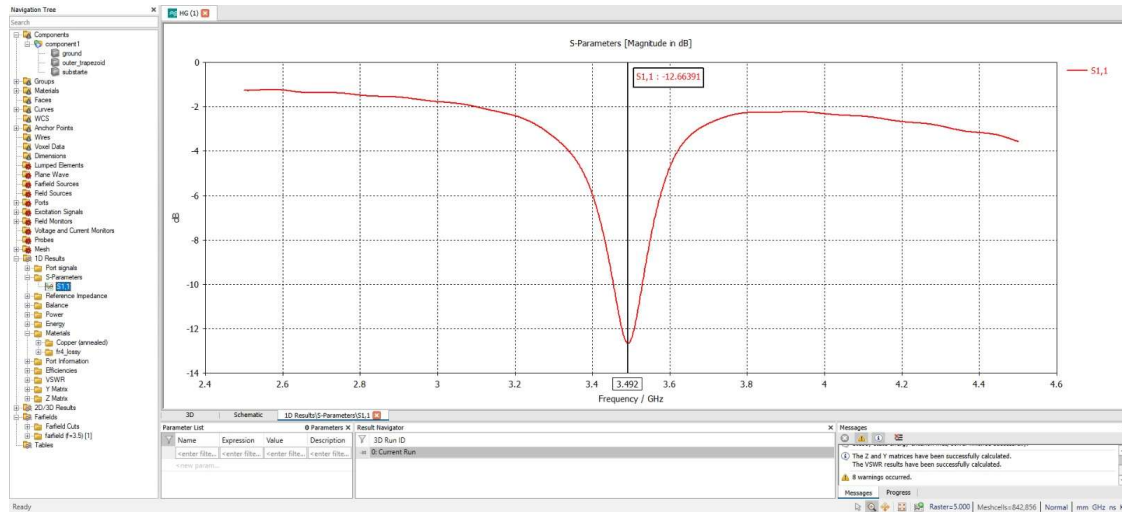


Figure 2. S-Parameters (Magnitude in dB) – Resonant frequency: 3.492 GHz, $S_{1,1}$: -12.664 dB

3.2 Voltage Standing Wave Ratio (VSWR)

The VSWR is a key parameter for evaluating impedance matching quality. A VSWR value between 1 and 2 is considered ideal for efficient power transfer. Figure 3 presents the VSWR characteristic of the proposed antenna. At the resonant frequency of 3.492 GHz, the VSWR measures 1.607, which falls within the acceptable range, confirming satisfactory matching between the feed and the radiating element.

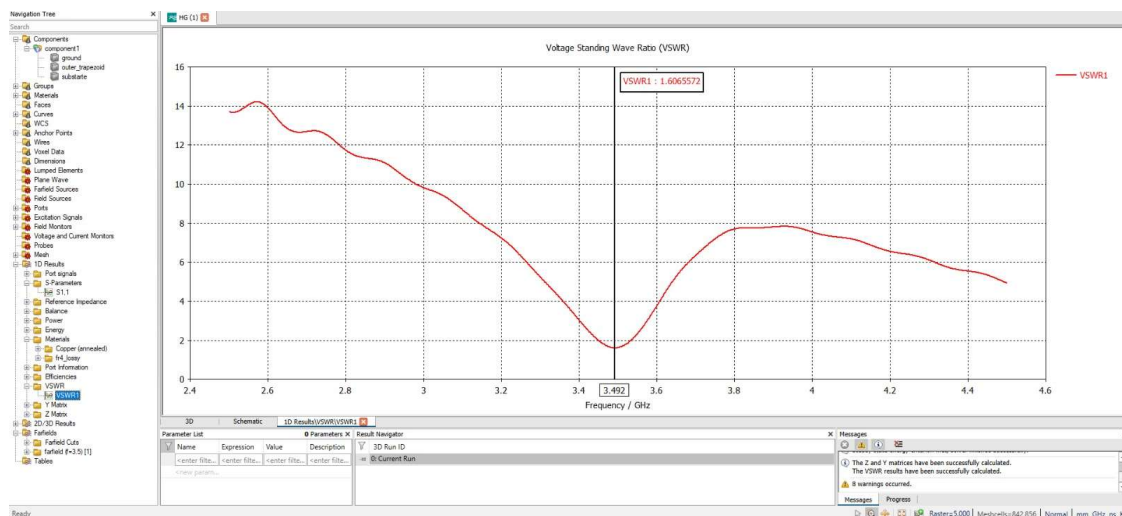


Figure 3. VSWR Characteristic of the Proposed Antenna – VSWR: 1.607 at 3.492 GHz

3.3 Smith Chart – Impedance Analysis

The Smith Chart in Figure 4 presents the input impedance of the antenna ($S_{1,1}$) in the impedance view over the frequency range of 2.4–4.6 GHz. The trajectory of the impedance

locus indicates that at 3.492 GHz the input impedance is well-matched near the center of the chart ($50\ \Omega$ normalized), consistent with the low return loss and VSWR observations. The impedance transitions from high reactive values at lower frequencies to a near-matched condition at resonance.

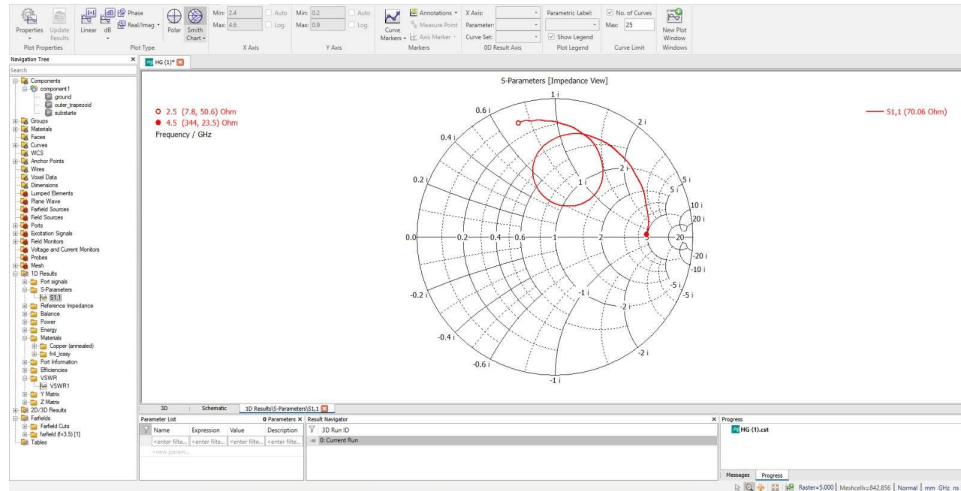


Figure 4. Smith Chart ($S_{1,1}$ Impedance View) – Reference impedance: $70.06\ \Omega$

3.4 Farfield Radiation Pattern (2D Polar)

The farfield directivity radiation pattern of the proposed antenna is analyzed at 3.5 GHz for the $\Phi = 90^\circ$ plane. Figures 5 and 6 present two views of the 2D polar farfield plot, showing the absolute directivity in dBi as a function of theta (elevation angle).

The key farfield parameters extracted from the simulation are as follows: the main lobe magnitude is 6.77 dBi directed at 0.0° , the angular width at -3 dB is 96.6° , and the side lobe level is -8.7 dB . The radiation pattern demonstrates a broadside radiation characteristic, which is typical for microstrip patch antennas.

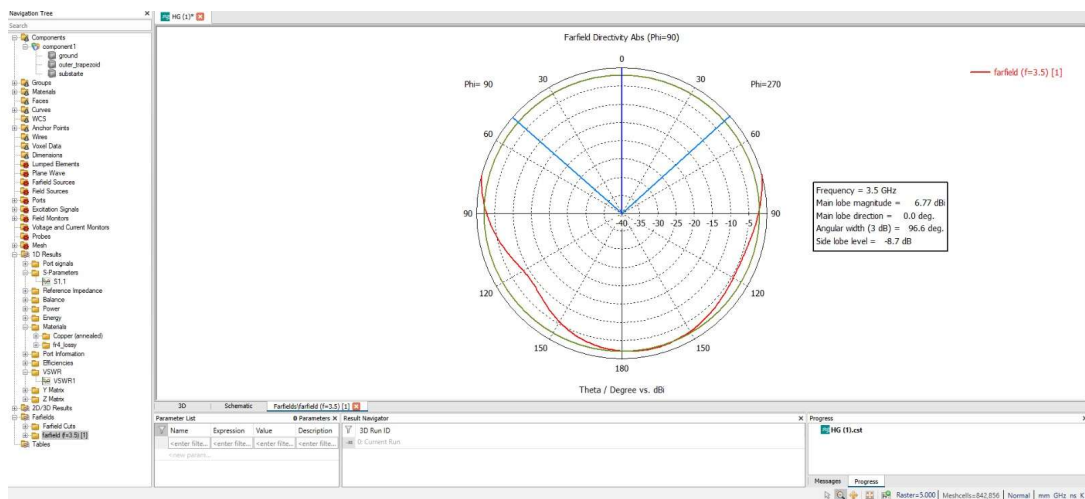


Figure 5. Farfield Directivity Abs (Phi=90) – Polar plot with outer normalized reference

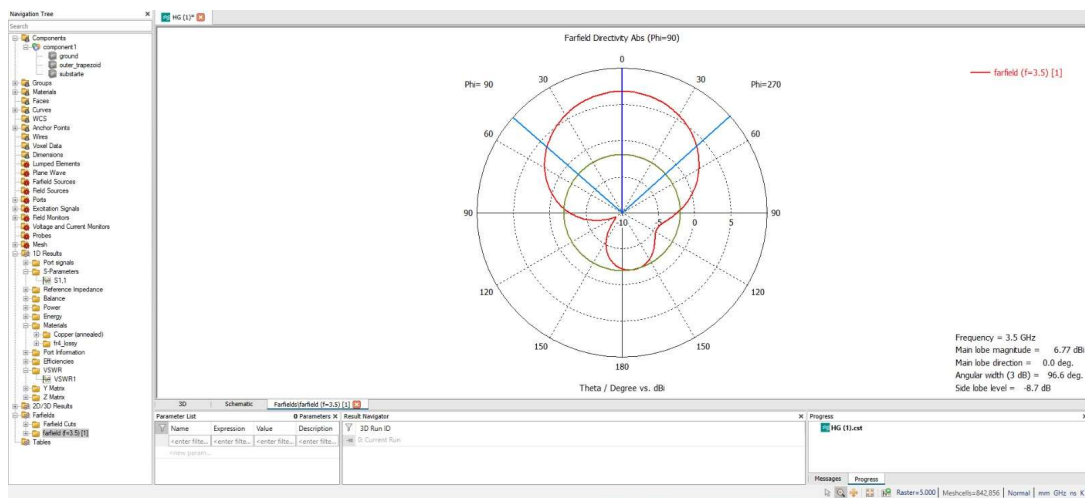


Figure 6. Farfield Directivity Abs (Phi=90) – Polar plot with dBi scale

3.5 3D Radiation Pattern

The three-dimensional farfield radiation pattern of the antenna at 3.5 GHz is presented in Figure 7. The 3D pattern clearly shows a hemisphere-dominant radiation behavior with maximum directivity in the broadside direction (toward 0°). The color scale on the right indicates directivity values ranging from approximately -33 dB (green) to +6.77 dBi (red/orange).

The farfield summary from the simulation panel confirms: Radiation Efficiency = -4.580 dB, Total Efficiency = -4.832 dB, and Directivity = 6.773 dBi.

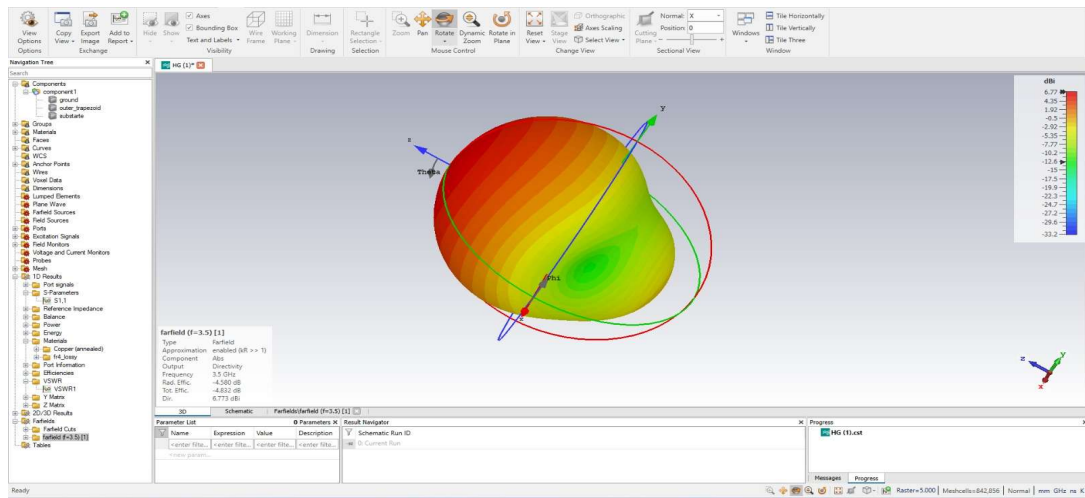


Figure 7. 3D Farfield Radiation Pattern at 3.5 GHz – Directivity: 6.773 dBi

IV. Results Summary and Comparison

Table 2 summarizes the key performance parameters obtained from the CST simulation and compares them with the reference values reported by Mukul Jain et al. (2017) for a similar trapezoid ring microstrip patch antenna design.

Parameter	Simulated Value	Reference Value (Mukul Jain et al.)
Resonant Frequency	3.492 GHz	3.498 GHz
Return Loss (S11)	-12.664 dB	-45.975 dB
VSWR	1.607	1.01
Main Lobe Directivity	6.77 dBi	6.107 dBi
Main Lobe Direction	0.0°	7.0°
Angular Width (3 dB)	96.6°	106.2°
Side Lobe Level	-8.7 dB	-14.2 dB

Table 2. Simulated Performance Parameters vs. Reference Values

The simulation results confirm that the antenna resonates at 3.492 GHz, which is within the target lower 5G band (3.4–3.6 GHz). The return loss of -12.664 dB and VSWR of 1.607 indicate acceptable impedance matching. The directivity of 6.77 dBi and broadside

radiation pattern are consistent with the expected behavior of a microstrip patch antenna. Minor deviations from the reference values are attributable to differences in meshing resolution (842,856 mesh cells in the current simulation), boundary conditions, and version-specific solver behavior in CST.

V. Applications

The trapezoid ring microstrip patch antenna designed in this study is well-suited for the following applications:

- 5G Sub-6 GHz Lower Band Communication (3.4–3.6 GHz) – suitable for outdoor base stations and user equipment in 5G NR deployments.
- Fixed Wireless Access (FWA) – enabling broadband connectivity over 5G networks in rural and semi-urban areas.
- IoT and Smart Device Integration – compact low-profile antenna suitable for embedding in smart meters, industrial IoT, and edge computing devices.
- WLAN and WiMAX Systems – the operating frequency overlaps with certain WiMAX frequency allocations, enabling dual-purpose usage.
- Microwave Engineering Research – serves as a compact demonstrator for evaluating novel patch geometries and feeding techniques.

VI. Conclusion

This report presents the design and CST Studio Suite simulation of a Trapezoid Ring Microstrip Patch Antenna for lower band 5G applications centered at 3.5 GHz. The antenna is fabricated on an FR4 substrate ($\epsilon_r = 4.3$, $\delta = 0.025$, $h = 1.6$ mm) and excited via a 50 Ω microstrip line feed with a patch insertion for impedance matching.

The simulation results demonstrate that the antenna achieves a resonant frequency of 3.492 GHz, a return loss of -12.664 dB, and a VSWR of 1.607 — all within acceptable limits for practical 5G deployment. The farfield analysis confirms a broadside radiation pattern with a main lobe directivity of 6.77 dBi, an angular beamwidth (3 dB) of 96.6° , and a side lobe level of -8.7 dB.

The antenna performance parameters are in reasonable agreement with the reference design, validating the simulation methodology. The compact form factor, ease of fabrication on standard PCB substrates, and compatibility with the 5G lower frequency band make this antenna a practical candidate for integration into next-generation

communication systems. Future work may include parametric optimization for improved return loss, gain enhancement using arrays, and fabrication with experimental validation.

References

- [1] Mukul Jain, Sarika Jain, "Design And Analysis Of Trapezoid Ring Microstrip Patch Antenna For Lower Band 5G Application," Int. Journal of Engineering Research and Application, ISSN: 2248-9622, Vol. 7, Issue 10, Part-5, pp. 15–18, October 2017.
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